Remote PPG heart-rate imaging

Estimating heart rate from ordinary video of skin

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Outline

- Introduction
- Basic concepts
- Remote PPG

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Introduction

- With the diffusion of camera-based DMS, new features are being developed to:
  - Improve drowsiness detection
  - Identify health problems
  - Identify intoxications (e.g., estimating the blood alcohol concentration)

- The new features are enabled by the capability of estimating the heart activity using a camera

- The method is known as remote photoplethysmography (rPPG)

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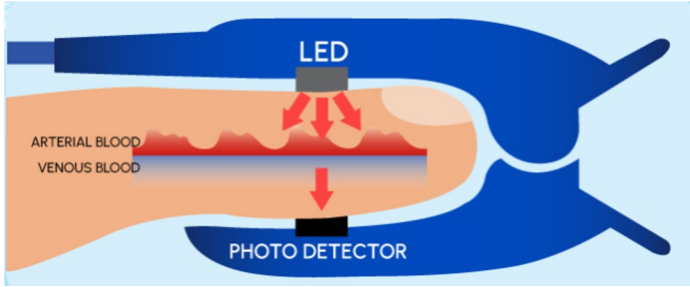
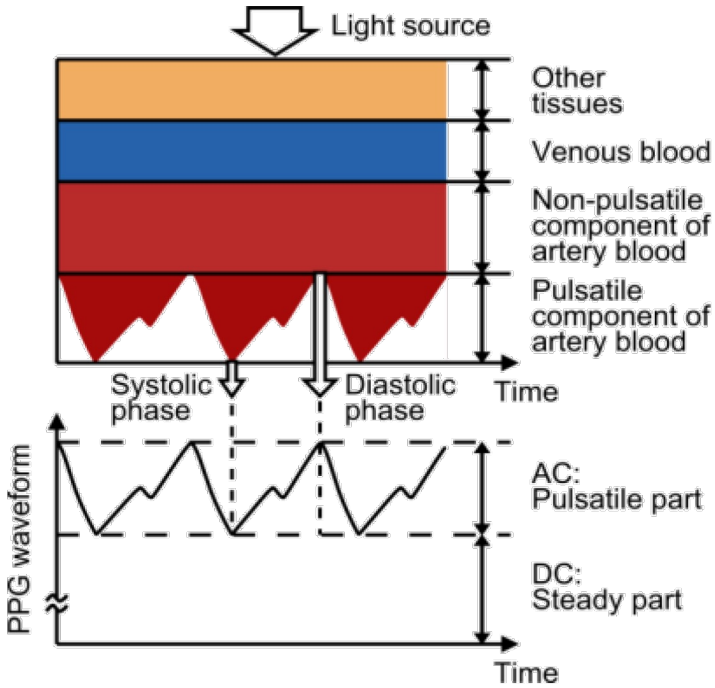
Basic concepts

- Photoplethysmography (PPG) is a non-invasive, low-cost optical technique that detects blood volume changes in the microvascular tissue bed using light from LEDs

- Operating Principle:
  - A light source (LED) illuminates the skin
  - a photodetector measures the intensity of light that is either transmitted or reflected
  - Blood absorbs more light than surrounding tissue
  - Fluctuations in blood volume during cardiac cycles change the detected light intensity, creating a PPG waveform





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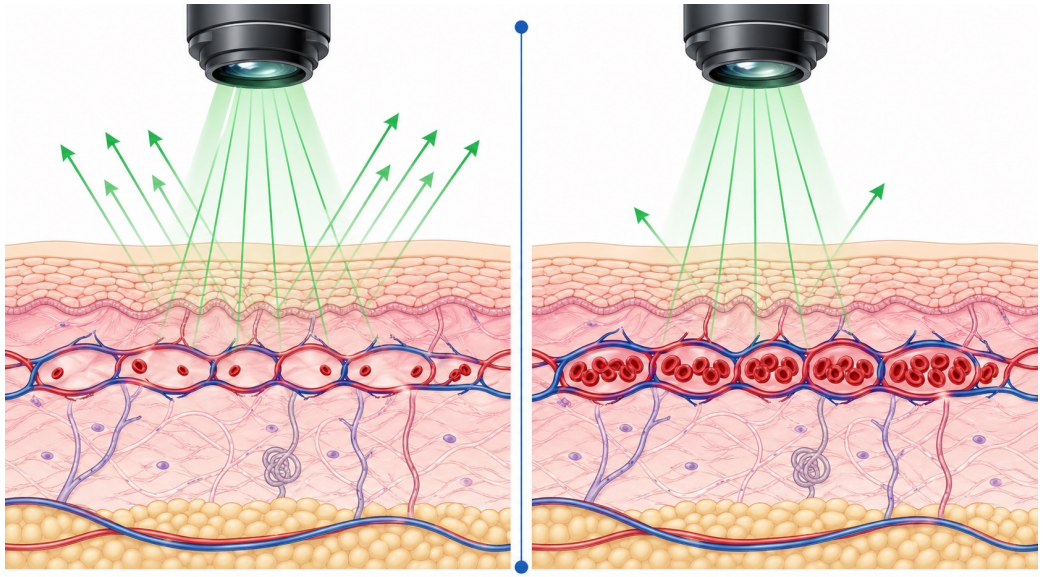
Remote PPG

- In rPPG, a camera is used to observe the skin of a subject
- Capillary blood volume changes with each heartbeat, modulating how much light returns from the skin to the camera
  - Lower perfusion: less hemoglobin in the illuminated capillary bed, less absorption, and more reflected light
  - Higher perfusion: more blood volume absorbs more light, so the reflected intensity measured by the camera drops
- The periodic intensity fluctuation is small
- Averaging many skin pixels and filtering around plausible heart-rate frequencies makes it measurable



Lower perfusion Higher perfusion

How remote PPG extracts heart rate

- A camera observe a subject
- The camera records skin reflectance fluctuations that are too subtle for human vision
- Reflected light varies with blood perfusion and tissue pulsation at the skin surface
- A video supplies RGB intensity time series for many skin pixels, frame by frame
- Filtering, Independent Component Analysis (ICA), FFT, and peak selection convert the dominant pulsatile frequency into beats per minute

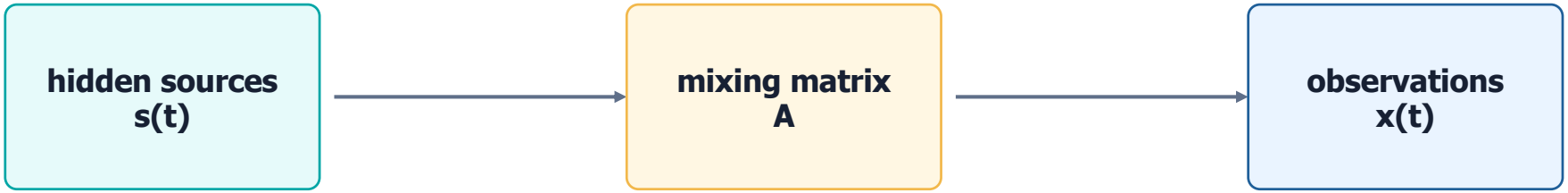
Independent Component Analysis (ICA)

- In rPPG the recorded RGB videos contain a mixture of:
  - Pulse signal
  - Motion
  - Illumination changes
  - Sensor noise

- ICA tries to separate these into distinct components

ICA

- Observed measurements are treated as unknown linear mixtures of hidden sources



$$s(t) = Ax(t)$$

- $x(t)$ is the observed signal over time: red channel, green channel, blue channel
- The goal of ICA is to estimate W such that $\hat{s}(t) = Wx(t)$
- The rows of W are filters that extract components from the observed mixtures

ICA

- ICA separates signals by exploiting statistical structure, not by knowing the physical source
  - Sources should not carry redundant information about each other
  - Pure sources usually look less Gaussian than their mixtures
  - Observed channels should be close to weighted sums of sources
  - Need at least as many mixtures as sources to recover all sources

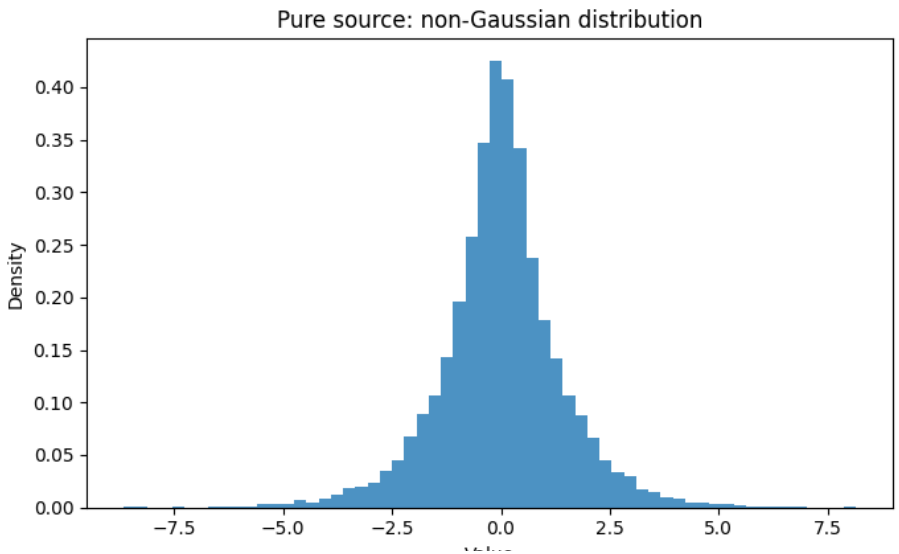
- Central limit intuition:
  - A weighted sum of independent sources tends to look more Gaussian than the original sources
  - ICA searches for directions that reverse this mixing

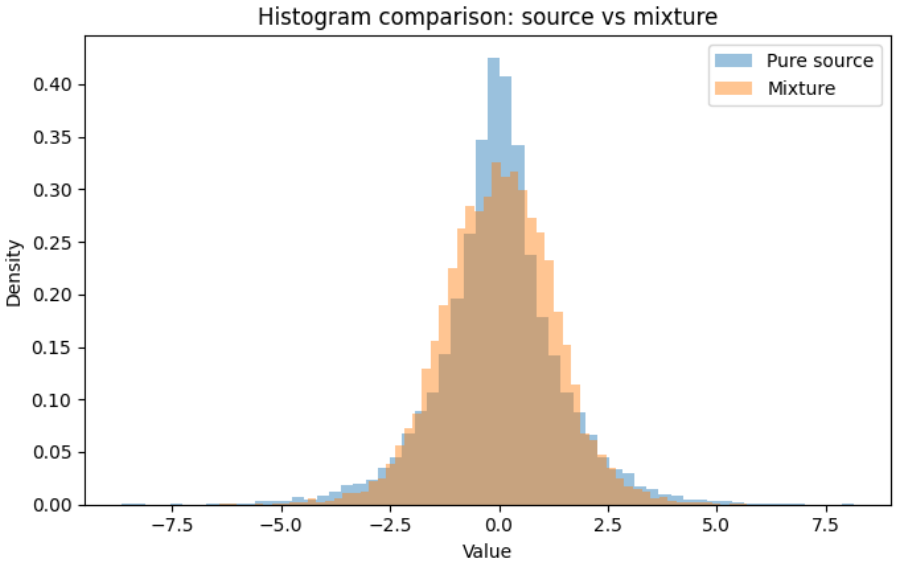
Central limit intuition

- When independent random signals are added together, the result tends to look more Gaussian (more bell-shaped, smoother, less "spiky" or structured) than the individual source signals

- Pure sources are the original underlying signals, before they get mixed together
  - Example: one person's voice, another person's voice, or a heartbeat signal plus a motion signal

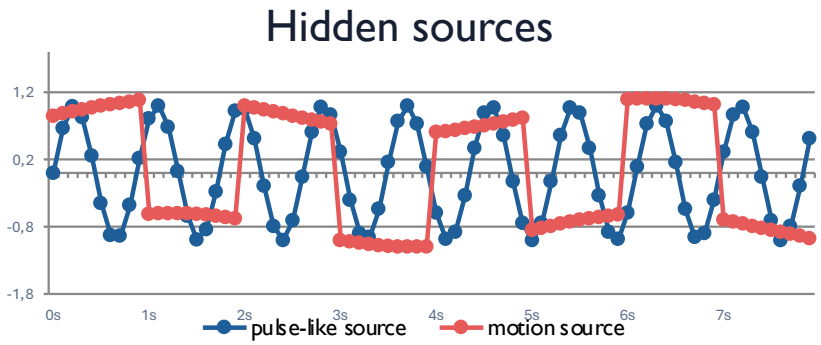
- A mixture is what you actually observe after several sources are added together

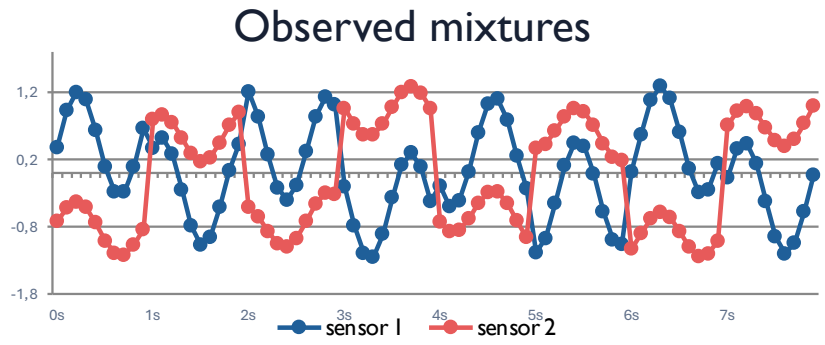




ICA

- ICA uses this fact to separate signals: It searches for projections of the observed data that are maximally non-Gaussian, because those projections are more likely to correspond to the original independent sources not to mixed combinations





Mixing matrix A
[0.80 0.45]
[0.35 -0.85]

ICA estimates W without knowing A or the source labels

Recovered components look like pulse and motion again

Post-processing identifies the component that matters

ICA algorithm

- Input: X observed signals, size = [m x N]
  - m = number of observed mixtures
  - N = number of samples
- Outputs:
  - S = estimated independent components
  - W = unmixing matrix

1
Center
$X = X - E(X)$

→

2
Whiten
$Z = V X$

→

3
Initialize
random w

→

4
Update
$w = E[z\, g(w^T z)] - E[g']w$

→

5
Normalize
$w = w / \|w\|$

→

6
Converge
$|w^T w_old| \approx 1$

Center the data

- To simplify the model and later computations, subtract the mean from each observed signal:

```
%-------------------------------------------------------
% 1. Center the data
%-------------------------------------------------------
meanX = mean(X, 2);                % mean of each row (signal)
Xc = X - meanX;                    % subtract mean from each signal
```

Whiten the data

- Transform the centered data so its components are uncorrelated and have unit variance
- This is usually done using eigenvalue decomposition of the covariance matrix

```
%-------------------------------------------------------
% 2. Whiten the data
%-------------------------------------------------------
C = cov(Xc');                          % covariance matrix, size [m x m]
[E, D] = eig(C);                       % eigenvectors and eigenvalues

% Whitening matrix
V = inv(sqrt(D)) * E';
Z = V * Xc;                            % whitened data
```

Initialize unmixing matrix

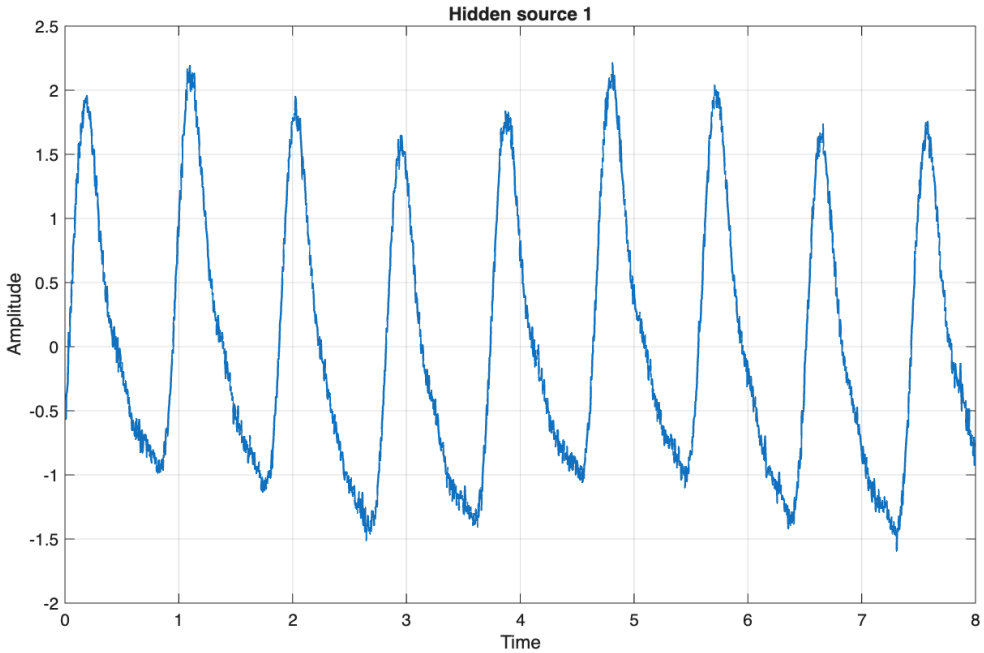
- The algorithm starts with a random unmixing matrix
- `num_components` is the number of components we want to decompose the observed signals

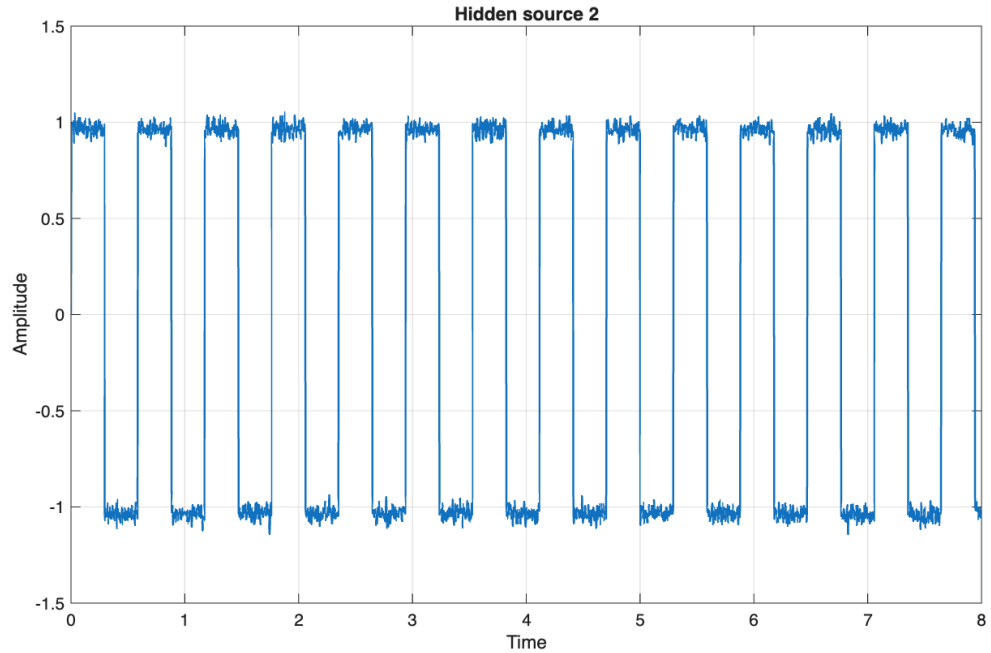
```
[m, N] = size(Z);
W = zeros(num_components, m);
```

Update

```
for p = 1:num_components
  w = randn(m, 1);
  w = w / norm(w);
  for iter = 1:max_iter
    w_old = w;
    y = w' * Z;
    g  = tanh(y);
    gp = 1 - g.^2;
    w = (Z * g') / N - mean(gp) * w_old;
    if p > 1
      w = w - W(1:p-1,:)' * (W(1:p-1,:) * w);
    end
    w = w / norm(w);
    % 6. Check convergence
    if abs(abs(w' * w_old) - 1) < tol
      break;
    end
   end
   % store converged component
   W(p, :) = w';
end
% 7. Recover independent components
S = W * Z;
```

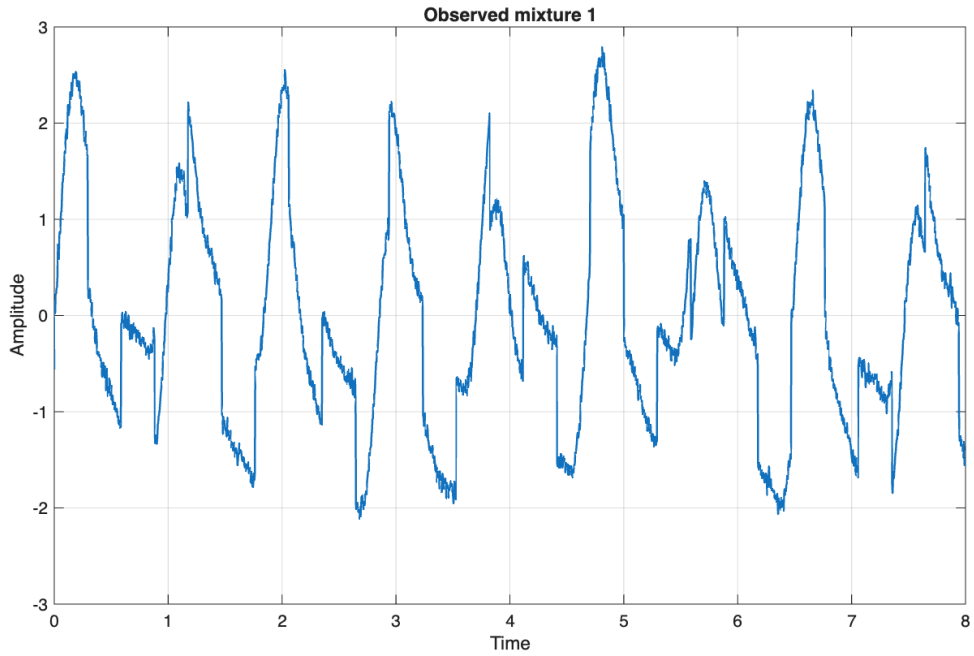
Example: hidden source

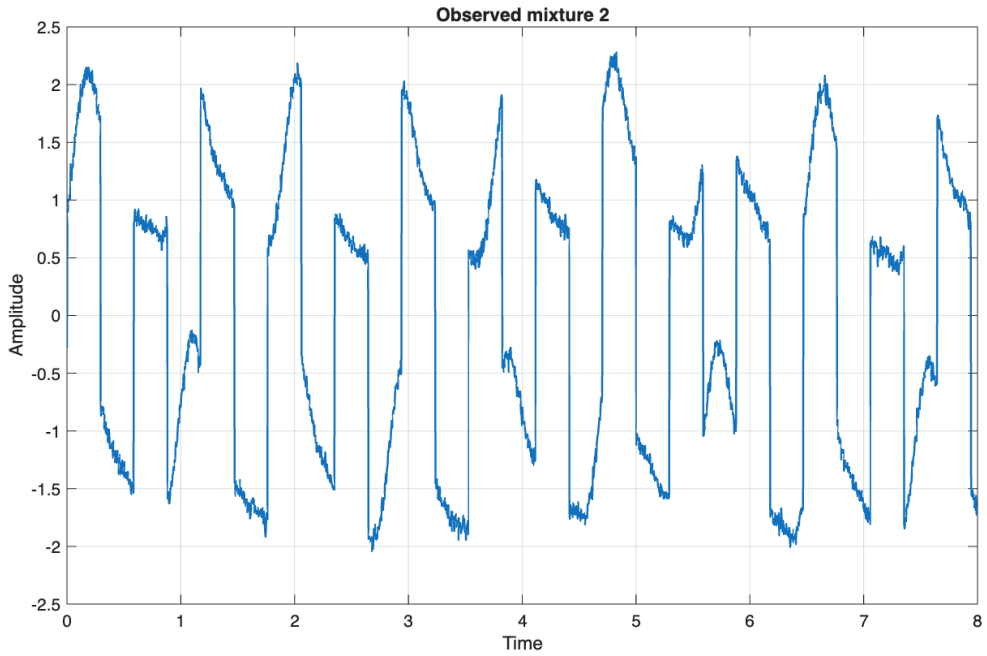




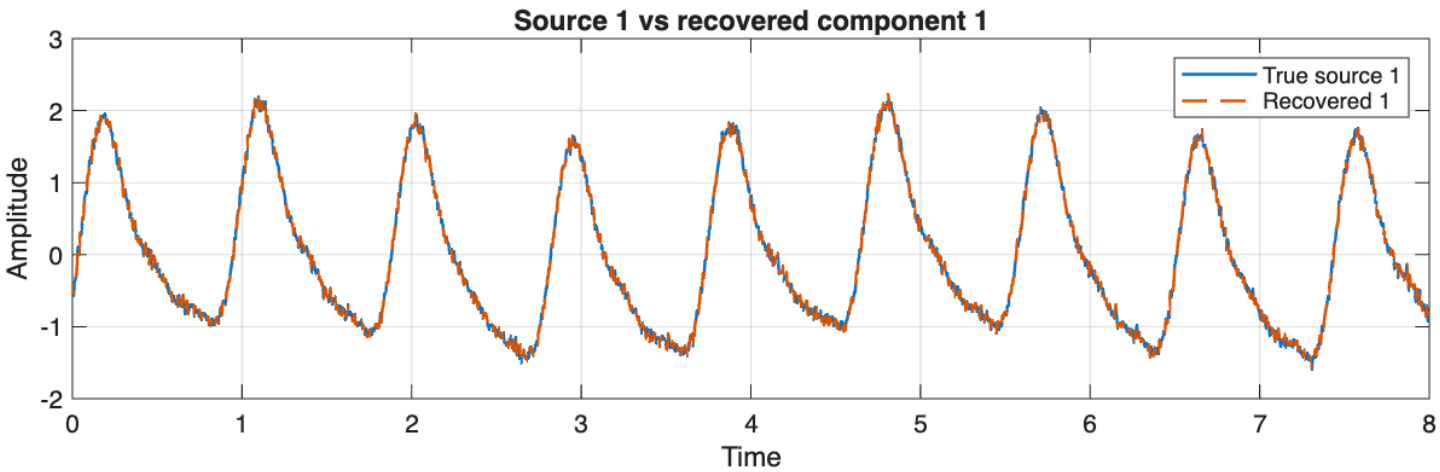
True mixing matrix: $A = \begin{bmatrix} 1.0 & 0.6 \\ 0.5 & 1.2 \end{bmatrix}$

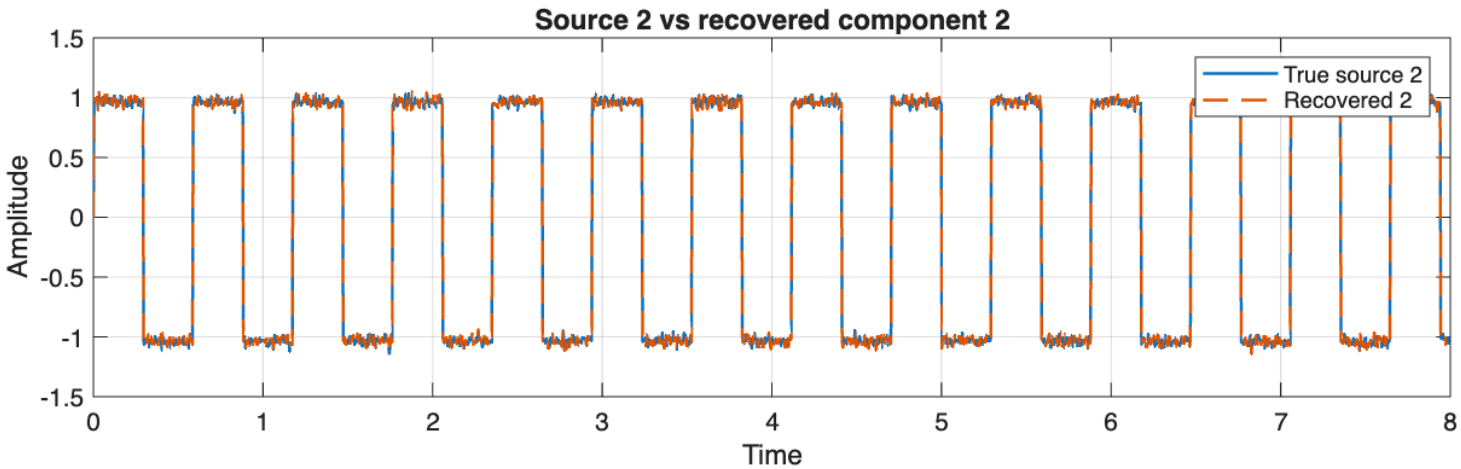
Example: observed samples





Example: recovered source vs hidden source





unmixing matrix

$$W = \begin{bmatrix} 0.9999 & -0.0120 \\ -0.0030 & 1.000 \end{bmatrix}$$

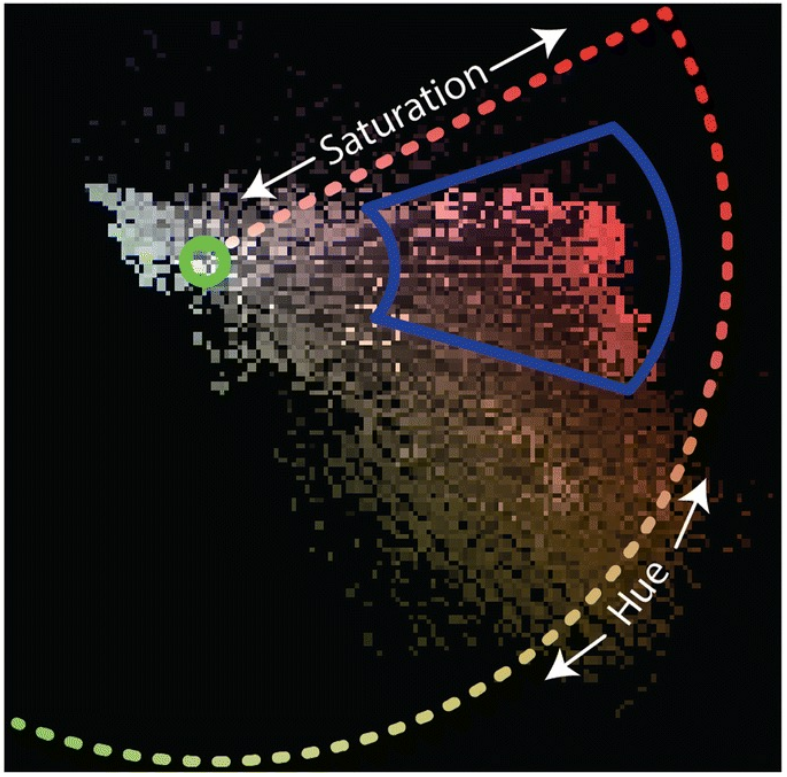
Algorithm pipeline

1 Crop video ROI > 2 Select skin pixels > 3 Average RGB traces > 4 Remove slow drift > 5 ICA signal separation > 6 FFT power spectrum > 7 Select HR peak

- Key idea: use many skin pixels to recover a faint pulsatile signal, then identify its dominant frequency
- Output: a single heart-rate estimate in beats per minute for the analyzed video segment

Skin-pixel selection

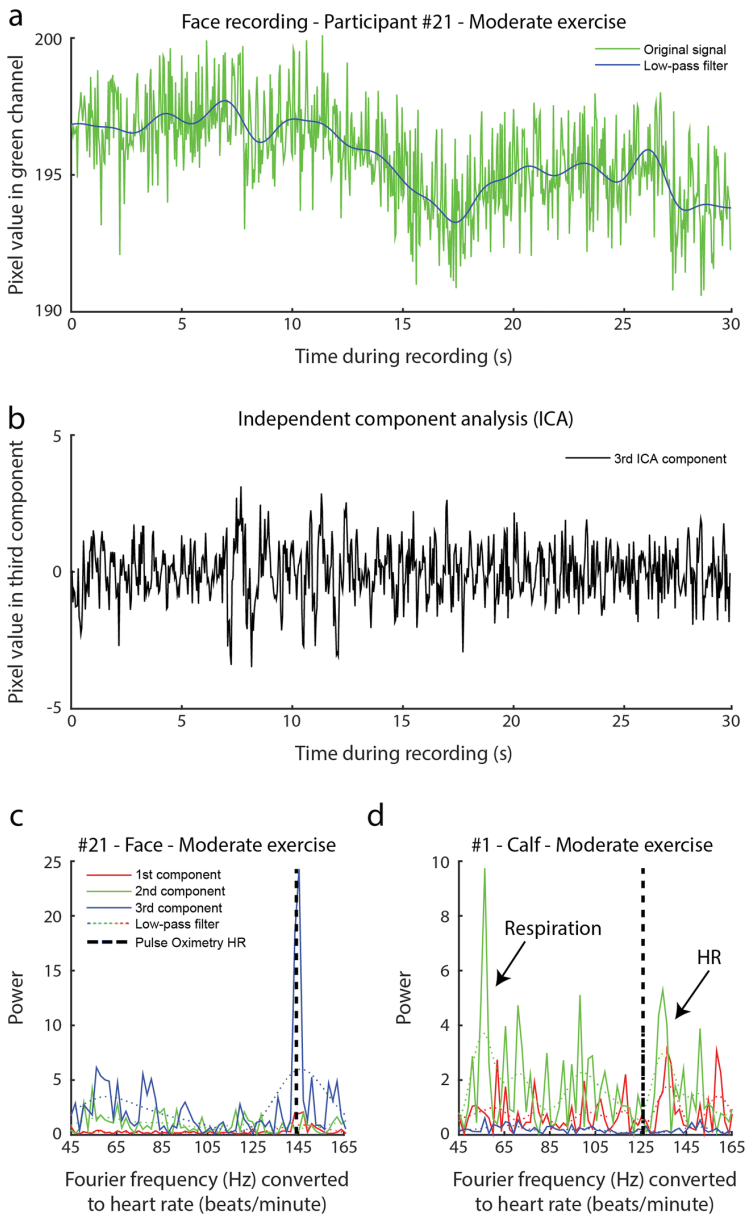
- Hue/saturation selection identifies skin-colored pixels; the selected pixels become the spatial support for rPPG signal extraction



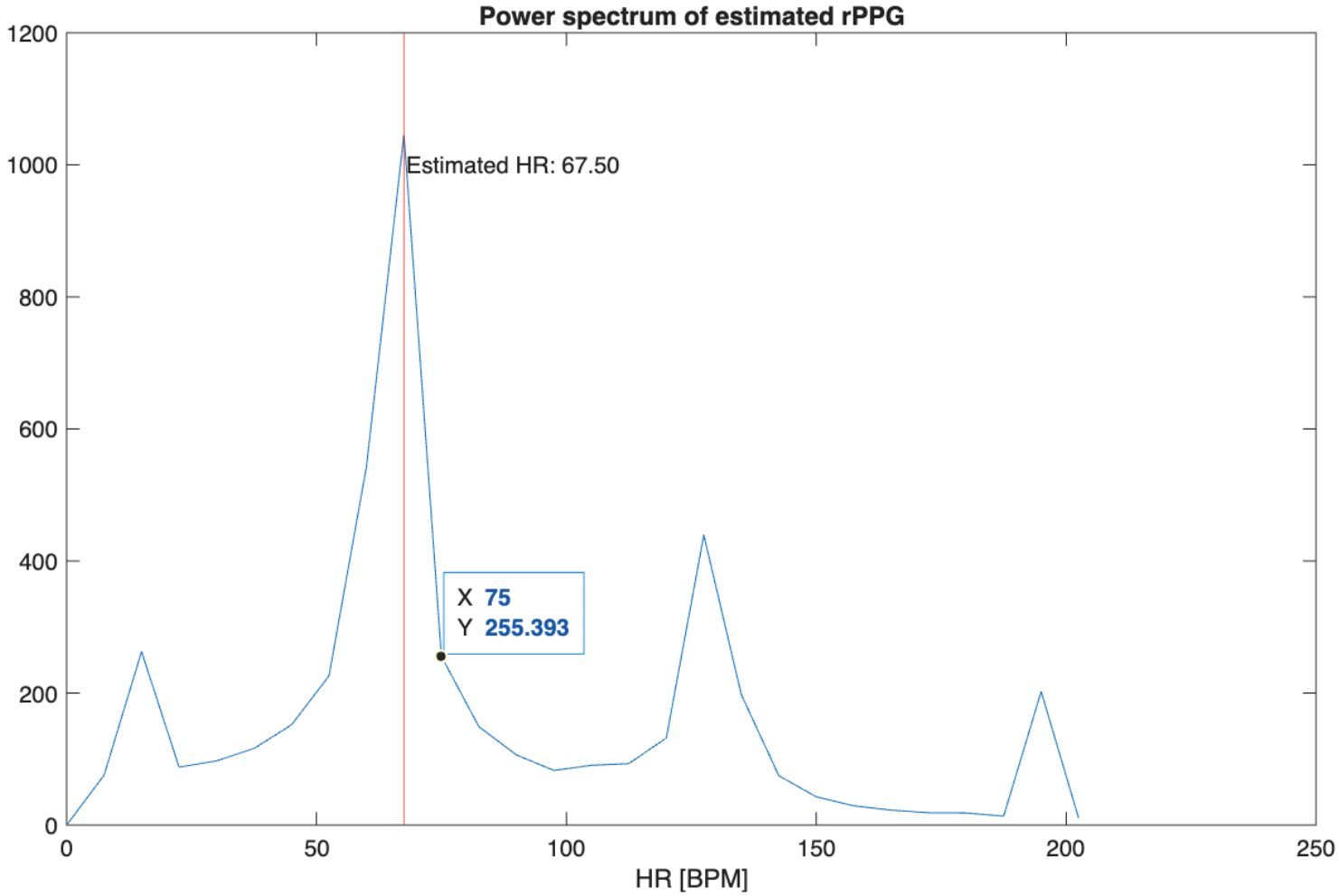
Peak selection

- The HR estimate depends on choosing the cardiac peak rather than motion or respiration

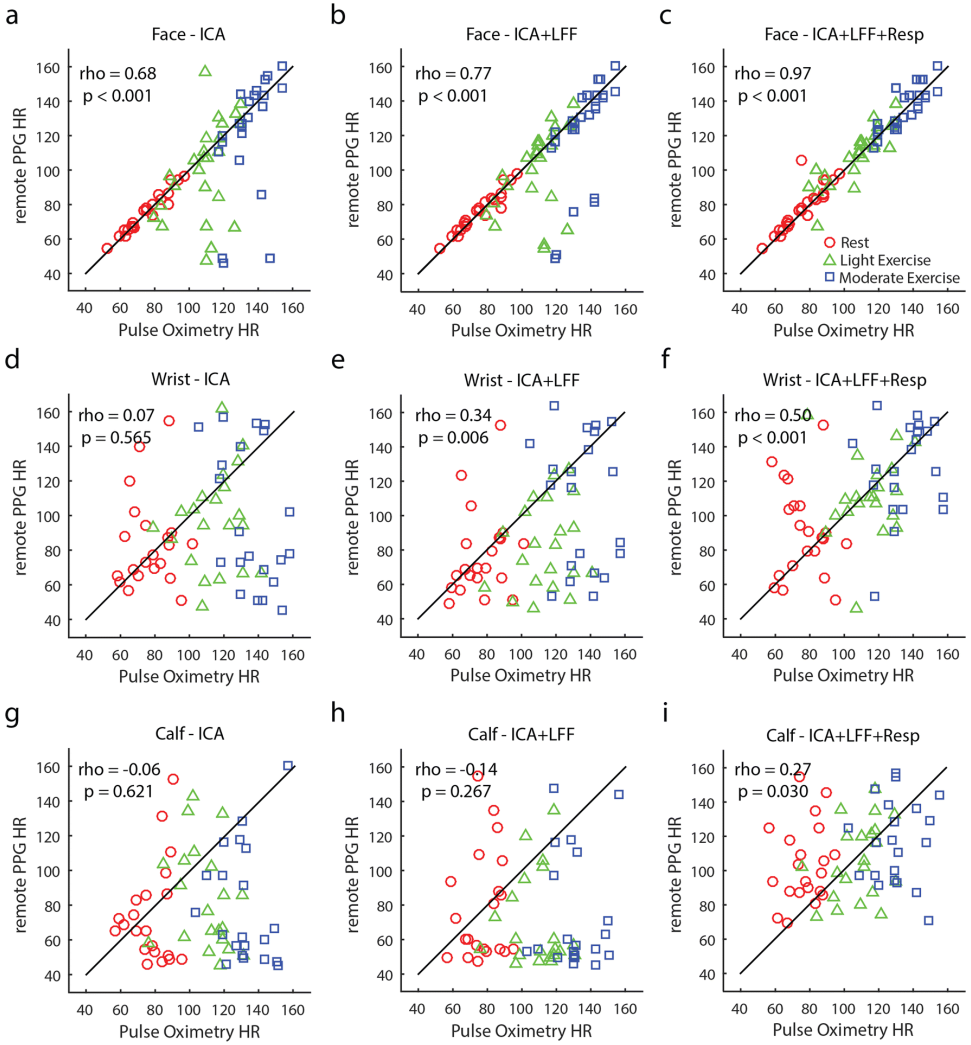
- The following operations are performed:
  - Transform ICA components into power spectra
  - Smooth spectra with a low-pass frequency filter
  - Select the highest power peak as HR unless a low-frequency respiration/movement peak is likely dominating



Example


Power spectrum of estimated rPPG
Estimated HR: 67.50
X 75
Y 255.393
1200
1000
800
600
400
200
0
0
50
100
150
200
250
HR [BPM]

Camera rPPG vs. pulse oximetry



- rPPG predictions are compared with pulse oximetry
- Different skin areas are considered:
  - Face: rho=0.97
  - Wrist: rho=0.50
  - Calf: rho=0.27
- Facial skin is the best areas for rPPG application